CARTRACK

Tong Seng Produce Pte Ltd: Elevating Delivery Reliability and Fleet Safety Across Singapore's Food Supply Chain



Location

Singapore

Industry

FMCG Producer and Distributor

Fleet Size

45+

www.cartrack.com

"Our customers trust us to deliver quality rice from our chilled warehouse to their tables. Cartrack gives us the visibility and control to honour that promise, every single delivery."

- William Ng, Operations Director at TSP

Business Overview

Tong Seng Produce Pte Ltd (TSP) is one of Singapore's most established distributors supplying more than 70 brands of essential grocery staples. Serving a diverse customer base spanning Singapore's supermarket chains, hotels, central kitchens, and catering operators, TSP operates in an industry where food safety, delivery reliability and operational efficiency are critical to maintaining customer trust and meeting service expectations. As part of its business transformation journey, TSP sought a smarter operational intelligence infrastructure to boost productivity and optimise its operations.

Challenges

Delivery Delays and Scheduling Disruptions

Besides delivering goods, TSP also provides value-added service by pouring rice into customers' kitchen bins. However, service time varies depending on kitchen accessibility and the number of bins to fill, making each stop difficult to predict. These delays cascaded across delivery routes, affecting subsequent arrival times, while management lacked visibility into which locations consistently caused overruns.

Speeding Risk and Driver Safety

To recover lost time, some drivers resorted to speeding between deliveries. As a company built on trust and reliability, unsafe driving behaviour not only put drivers and other road users at risk but also undermined TSP's brand reputation.

Rising Fuel Cost and Limited Carbon Emission Data

Management observed rising fuel consumption across the fleet but lacked trip-level insights to identify the root causes. Manual fuel records also made it difficult to accurately track Scope 1 and 2 carbon emissions, which require more granular fuel consumption data.

Reactive Maintenance and Unplanned Downtime

Without real-time visibility into vehicle health and engine diagnostics, management could only respond after breakdowns occurred, increasing maintenance costs and disrupting service reliability and customer relationships.

High Administrative Workload

With deliveries spanning multiple customer locations, operations managers lacked real-time visibility into vehicle locations or delivery progress. Significant time was spent calling drivers for updates, creating inefficiencies that became harder to manage as the business expanded.

Solutions Implementation

1. Geofences to Measure Dwell Time and Improve Route Planning

Geofences were deployed at every delivery location to monitor arrival and departure times, dwell time, and visit frequency. These insights gave managers visibility of where time was spent, enabling better route planning and buffers for time-sensitive deliveries.

2. Driver ID Tag to Ensure Individual Accountability

Driver ID (DID) tags linked every trip, speeding event, and idling incident to an individual driver, eliminating ambiguity and ensuring fair, consistent accountability. With accurate driver attribution, performance management became more transparent and effective.

3. Speed Limiters to Enhance Driver Safety

The installation of speed limiters provided both preventive control and real-time monitoring of speeding events, which fed directly into the driver scorecard system for coaching. Since implementation, speeding incidents have reduced significantly, helping create a safer fleet while reducing operational risk.

4. Scorecard System to Encourage Behavioural Change

With the scorecard system, identifying high-risk drivers has now shifted from reactive to proactive. All drivers are now ranked, and those below the benchmark automatically trigger a report to managers, signifying the need for targeted coaching. Performance incentives are also tied to scorecard outcomes, rewarding safe and fuel-efficient drivers while fostering healthy competition and measurable behavioural improvement.

5. Fuel Sensors to Improve Fuel Efficiency and Emissions Reporting

Fuel sensors provided real-time, trip-level fuel consumption data, enabling managers to analyse fuel usage by vehicle, driver, and trip while distinguishing idling-related wastage from in-transit inefficiencies. This data now supports TSP's Scope 1 and 2 carbon emissions tracking, providing the auditable insights needed to support the company's reduction targets.

6. CANbus to Enable Preventive Maintenance

CANbus connectivity provided real-time engine diagnostics, including engine hours, mileage, and tyre health. Maintenance alerts based on actual vehicle usage enabled servicing before breakdowns occurred, reducing unplanned downtime and ensuring fleet reliability for time-critical deliveries.

7. GPS Tracking for Real-Time Fleet Visibility

Cartrack's GPS solution gave TSP's operations team live visibility of every vehicle across the island, eliminating the need for phone-based driver check-ins. For an operations team managing concurrent deliveries across multiple locations with varying scheduling requirements, having all fleet movements displayed on a single dashboard allowed them to make faster, better-informed operational decisions.

Business Impact

11% Improvement in Fleet Safety Performance

through reduced harsh events across the fleet

- 13% Reduction in Speeding
- 14% Reduction in Aggressive Acceleration
- 20% Reduction in Sudden Braking
- 10% Reduction in Harsh Cornering

10% Improvement in On-Time Delivery Performance

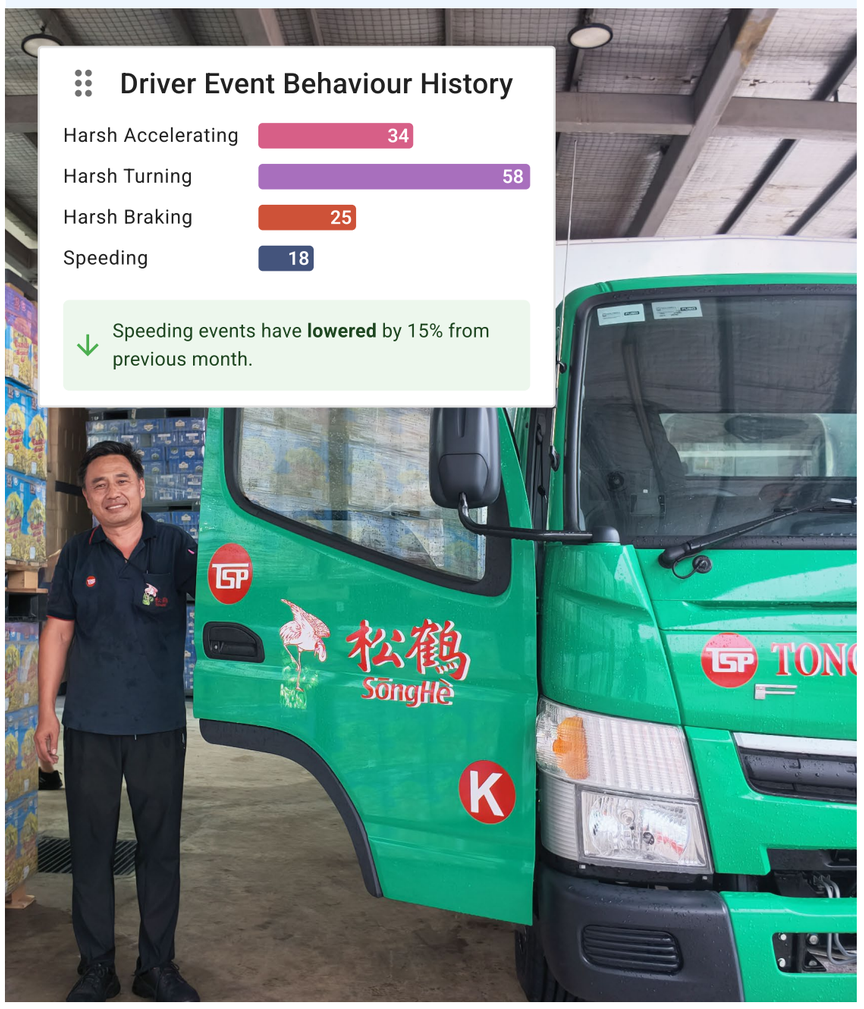
through smarter route sequencing using geofence dwell-time insights

27% Improvement in Fuel Efficiency

achieved through targeted reduction of speeding and idling behaviour

7% Improvement in Driver Score

via a shift from reactive to proactive behavioural management and coaching



Conclusion

By strengthening fleet visibility, improving accountability, and taking a more proactive approach to fleet operations, TSP has built a safer, more efficient, and more reliable food distribution operation. Cartrack supports this transformation by providing the operational intelligence needed to make faster and better-informed decisions every day.

Ready to build a fleet your customers can trust?

www.cartrack.com

CARTRACK